
Engineering Mathematics – III

Two-Mark Questions & Answers

Part – A / Question Bank

80 Questions • 4 Units • 20 Q&A per Unit

Laplace Transforms | Analytic Functions

Complex Integration | Fourier Series

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A consolidated revision aid for university Part–A examinations.

UNIT I – LAPLACE TRANSFORMS

Transforms, inverse transforms, theorems and applications

Q1. Define the Laplace transform.

Ans: The Laplace transform of a function $f(t)$ defined for $t \geq 0$ is $L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$, provided the integral converges. Here s is a parameter (real or complex).

Q2. State the sufficient conditions for the existence of the Laplace transform.

Ans: $L\{f(t)\}$ exists if $f(t)$ is (i) piecewise (sectionally) continuous on every finite interval in $[0, \infty)$, and (ii) of exponential order, i.e. $|f(t)| \leq Me^{\alpha t}$ for some constants $M > 0$, α and large t .

Q3. State the initial value theorem.

Ans: If $L\{f(t)\} = F(s)$, then $\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} sF(s)$, provided the limits exist.

Q4. State the final value theorem.

Ans: If $L\{f(t)\} = F(s)$, then $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$, provided $sF(s)$ has no poles in the right half-plane or on the imaginary axis.

Q5. State the convolution theorem on Laplace transforms.

Ans: If $L\{f(t)\} = F(s)$ and $L\{g(t)\} = G(s)$, then $L\{(f * g)(t)\} = F(s)G(s)$, where $(f * g)(t) = \int_0^t f(u)g(t-u) du$.

Q6. State the Laplace transform of a periodic function.

Ans: If $f(t)$ is periodic with period T ($f(t+T) = f(t)$), then $L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$.

Q7. Find $L(\cos^2 t)$.

Ans: Using $\cos^2 t = \frac{1}{2}(1 + \cos 2t)$, $L(\cos^2 t) = \frac{1}{2} \left[\frac{1}{s} + \frac{s}{s^2 + 4} \right]$.

Q8. Find $L(\sin 3t \cos t)$.

Ans: Since $\sin 3t \cos t = \frac{1}{2}(\sin 4t + \sin 2t)$, $L(\sin 3t \cos t) = \frac{2}{s^2 + 16} + \frac{1}{s^2 + 4}$.

Q9. Find $L^{-1} \left[\frac{s}{(s+1)^2 + 1} \right]$.

Ans: Writing $s = (s+1) - 1$, $L^{-1} \left[\frac{(s+1) - 1}{(s+1)^2 + 1} \right] = e^{-t} \cos t - e^{-t} \sin t$.

Q10. Find $L^{-1} \left[\frac{s}{(s+2)^2} \right]$.

Ans: Since $\frac{s}{(s+2)^2} = \frac{1}{s+2} - \frac{2}{(s+2)^2}$, $L^{-1} \left[\frac{s}{(s+2)^2} \right] = e^{-2t}(1 - 2t)$.

Q11. Find $L^{-1} \left[\frac{s+2}{s^2 + 2s + 5} \right]$.

Ans: As $s^2 + 2s + 5 = (s+1)^2 + 4$ and $s+2 = (s+1) + 1$, $L^{-1} = e^{-t} \cos 2t + \frac{1}{2} e^{-t} \sin 2t$.

Q12. Find $L\left[\frac{\sin t}{t}\right]$.

Ans: Using $L\{f(t)/t\} = \int_s^\infty F(u) du$ with $L(\sin t) = \frac{1}{u^2 + 1}$, $L\left[\frac{\sin t}{t}\right] = \int_s^\infty \frac{du}{u^2 + 1} = \frac{\pi}{2} - \tan^{-1} s = \cot^{-1} s$.

Q13. Find the Laplace transform of $\frac{1+2t}{\sqrt{t}}$.

Ans: Writing it as $t^{-1/2} + 2t^{1/2}$ and using $L(t^n) = \frac{\Gamma(n+1)}{s^{n+1}}$, $L\left[\frac{1+2t}{\sqrt{t}}\right] = \sqrt{\frac{\pi}{s}} + \frac{\sqrt{\pi}}{s^{3/2}}$.

Q14. Find the Laplace transform of $(t-1)^2 u(t-1)$.

Ans: By the second shifting theorem with $g(t) = t^2$, $G(s) = \frac{2}{s^3}$: $L\{(t-1)^2 u(t-1)\} = e^{-s} \frac{2}{s^3} = \frac{2e^{-s}}{s^3}$.

Q15. Find $L\{e^{3t} \sin 2t \sin t\}$.

Ans: Since $\sin 2t \sin t = \frac{1}{2}(\cos t - \cos 3t)$, then by first shifting ($s \rightarrow s - 3$): $\frac{1}{2} \left[\frac{s-3}{(s-3)^2 + 1} - \frac{s-3}{(s-3)^2 + 9} \right]$.

Q16. Evaluate $\int_0^\infty t e^{-2t} dt$ using Laplace transforms.

Ans: This is $L(t)$ evaluated at $s = 2$. Since $L(t) = \frac{1}{s^2}$, $\int_0^\infty t e^{-2t} dt = \frac{1}{2^2} = \frac{1}{4}$.

Q17. Give the Laplace transform of the Dirac delta function.

Ans: $L\{\delta(t)\} = 1$, and more generally $L\{\delta(t-a)\} = e^{-as}$ for $a \geq 0$.

Q18. Find $f(0)$ if $L[f(t)] = \frac{1}{s(s+1)}$.

Ans: By the initial value theorem, $f(0) = \lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} \frac{1}{s+1} = 0$.

Q19. State the first shifting (translation) theorem.

Ans: If $L\{f(t)\} = F(s)$, then $L\{e^{at} f(t)\} = F(s-a)$. Conversely $L^{-1}\{F(s-a)\} = e^{at} f(t)$.

Q20. Find $L^{-1}\left[\frac{1}{s^2 + 4s + 13}\right]$.

Ans: Since $s^2 + 4s + 13 = (s+2)^2 + 9$, $L^{-1}\left[\frac{1}{(s+2)^2 + 3^2}\right] = \frac{1}{3} e^{-2t} \sin 3t$.

UNIT II – ANALYTIC FUNCTIONS

Cauchy–Riemann equations, harmonic functions, conformal and Möbius mappings

Q21. Define an analytic function.

Ans: A function $f(z)$ is analytic at a point z_0 if it is differentiable at z_0 and at every point of some neighbourhood of z_0 . A function analytic at every point of a region is analytic in that region.

Q22. State the Cauchy–Riemann (C–R) equations.

Ans: If $f(z) = u(x, y) + i v(x, y)$ is analytic, then $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$, i.e. $u_x = v_y$, $u_y = -v_x$.

Q23. Write the Cauchy–Riemann equations in polar coordinates.

Ans: $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$ and $\frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}$.

Q24. State any two properties of an analytic function.

Ans: (i) The real and imaginary parts of an analytic function are harmonic (satisfy Laplace's equation). (ii) An analytic function is infinitely differentiable; an analytic function with constant modulus (or constant real / imaginary part) is constant.

Q25. Define a harmonic function.

Ans: A real function $\phi(x, y)$ with continuous second-order partial derivatives is harmonic if it satisfies Laplace's equation $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$.

Q26. Define the harmonic conjugate of a function.

Ans: If u and v are harmonic and together satisfy the C–R equations so that $f = u + iv$ is analytic, then v is called the harmonic conjugate of u .

Q27. Show that $f(z) = \cos z$ is analytic.

Ans: Here $u = \cos x \cosh y$, $v = -\sin x \sinh y$. Then $u_x = -\sin x \cosh y = v_y$ and $u_y = \cos x \sinh y = -v_x$. The C–R equations hold with continuous partials everywhere, so $\cos z$ is analytic (entire).

Q28. Verify whether $f(z) = \sin z$ is analytic.

Ans: With $u = \sin x \cosh y$, $v = \cos x \sinh y$: $u_x = \cos x \cosh y = v_y$ and $u_y = \sin x \sinh y = -v_x$. C–R equations are satisfied everywhere, so $\sin z$ is analytic (entire).

Q29. Show that $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic.

Ans: $u_x = 6xy + 4x \Rightarrow u_{xx} = 6y + 4$; $u_y = 3x^2 - 3y^2 - 4y \Rightarrow u_{yy} = -6y - 4$. Hence $u_{xx} + u_{yy} = (6y + 4) + (-6y - 4) = 0$, so u is harmonic.

Q30. Show that $v = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 4xy$ satisfies Laplace's equation.

Ans: $v_{xx} = -\sin x \cosh y - 2 \cos x \sinh y + 2$ and $v_{yy} = \sin x \cosh y + 2 \cos x \sinh y - 2$. Adding gives $v_{xx} + v_{yy} = 0$, so v is harmonic.

Q31. Find the analytic function whose real part is $u = y + e^x \cos y$.

Ans: By Milne–Thomson's method, $f'(z) = u_x - i u_y = e^x \cos y - i(1 - e^x \sin y)$. Putting $x = z$, $y = 0$: $f'(z) = e^z - i$. Integrating, $f(z) = e^z - iz + c$.

Q32. Find the value of k if $u = kx^2 + y^2$ is the real part of an analytic function.

Ans: u must be harmonic: $u_{xx} + u_{yy} = 2k + 2 = 0 \Rightarrow k = -1$.

Q33. Show that an analytic function with constant modulus is constant.

Ans: If $|f| = c$. For $c = 0$, $f \equiv 0$. For $c \neq 0$, $u^2 + v^2 = c^2$; differentiating and using the C–R equations ($u_y = -v_x$, $v_y = u_x$) gives $u_x = v_x = 0$. Thus $f'(z) = 0$, so f is constant.

Q34. Prove that an analytic function with constant imaginary part is constant.

Ans: If $v = \text{constant}$, then $v_x = v_y = 0$. By C–R, $u_x = v_y = 0$ and $u_y = -v_x = 0$, so u is also constant. Hence $f = u + iv$ is constant.

Q35. Define conformal mapping.

Ans: A mapping $w = f(z)$ that preserves angles in both magnitude and sense between any two oriented curves through a point is called conformal. It is conformal at z_0 if f is analytic there and $f'(z_0) \neq 0$.

Q36. Define a critical point of a transformation.

Ans: A point at which $f'(z) = 0$ is called a critical point of the transformation $w = f(z)$. Conformality (angle preservation) fails at a critical point.

Q37. Define a Möbius (bilinear) transformation.

Ans: A transformation of the form $w = \frac{az + b}{cz + d}$, where a, b, c, d are complex constants with $ad - bc \neq 0$, is called a Möbius (bilinear / linear fractional) transformation.

Q38. Find the fixed points of $w = \frac{z - 1}{z + 1}$.

Ans: Set $z = \frac{z - 1}{z + 1} \Rightarrow z(z + 1) = z - 1 \Rightarrow z^2 = -1 \Rightarrow z = \pm i$. Fixed points: $i, -i$.

Q39. Find the fixed points of $w = \frac{1 - z}{1 + z}$.

Ans: Set $z = \frac{1 - z}{1 + z} \Rightarrow z^2 + 2z - 1 = 0 \Rightarrow z = -1 \pm \sqrt{2}$.

Q40. Find the fixed points of $w = \frac{2zi + 5}{z - 4i}$.

Ans: Set $z = \frac{2zi + 5}{z - 4i} \Rightarrow z^2 - 6iz - 5 = 0 \Rightarrow z = \frac{6i \pm 4i}{2}$. Fixed points: $z = i$ and $z = 5i$.

UNIT III – COMPLEX INTEGRATION

Cauchy's theorems, singularities, residues and Laurent series

Q41. State Cauchy's integral theorem (fundamental theorem).

Ans: If $f(z)$ is analytic and $f'(z)$ is continuous at all points inside and on a simple closed curve C , then $\oint_C f(z) dz = 0$.

Q42. State Cauchy's integral formula.

Ans: If $f(z)$ is analytic inside and on a simple closed curve C and a lies inside C , then $f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - a} dz$, and $f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - a)^{n+1}} dz$.

Q43. Evaluate $\oint_C \frac{dz}{z - 2}$ where C is the circle with centre 2 and radius 4.

Ans: The point $z = 2$ lies inside C . By Cauchy's integral formula with $f(z) = 1$, $\oint_C \frac{dz}{z - 2} = 2\pi i$.

Q44. Evaluate $\oint_C \frac{3z^2 + 7z + 1}{z + 1} dz$ where $C: |z| = \frac{1}{2}$.

Ans: The pole $z = -1$ lies *outside* C (since $|-1| = 1 > \frac{1}{2}$). As the integrand is analytic inside C , the integral = 0.

Q45. Evaluate $\oint_C \frac{dz}{2z + 1}$ where $C: |z| = \sqrt{2}$.

Ans: The pole $z = -\frac{1}{2}$ lies inside C . Writing $\frac{1}{2z + 1} = \frac{1}{2} \cdot \frac{1}{z + \frac{1}{2}}$, $\oint_C \frac{dz}{2z + 1} = \frac{1}{2}(2\pi i) = \pi i$.

Q46. Define a singular point (isolated singularity).

Ans: A point z_0 at which $f(z)$ ceases to be analytic is a singular point. If f is analytic in a deleted neighbourhood $0 < |z - z_0| < \rho$ but not at z_0 , then z_0 is an isolated singularity.

Q47. Define a pole of order n .

Ans: If the principal part of the Laurent expansion of f about z_0 has a finite number of terms, the highest negative power being $(z - z_0)^{-n}$, then z_0 is a pole of order n ($n = 1$ gives a simple pole).

Q48. Define an essential singularity.

Ans: If the principal part of the Laurent expansion of f about z_0 contains an infinite number of terms, then z_0 is an essential singularity.

Q49. Define a removable singularity.

Ans: If the principal part of the Laurent expansion about z_0 has no terms (all negative-power coefficients vanish), then z_0 is a removable singularity; f can be redefined at z_0 to be analytic there.

Q50. Find the singular points of $f(z) = \frac{1}{z(e^z - 1)}$.

Ans: The denominator vanishes when $z = 0$ or $e^z = 1 \Rightarrow z = 2n\pi i$. Thus $z = 0$ is a pole of order 2, and $z = \pm 2\pi i, \pm 4\pi i, \dots$ are simple poles. Singular points: $z = 2n\pi i, n \in \mathbb{Z}$.

Q51. Find the residue of $f(z) = \frac{e^{2z}}{(z+1)^2}$ at its pole.

Ans: $z = -1$ is a pole of order 2. $\text{Res} = \lim_{z \rightarrow -1} \frac{d}{dz} [e^{2z}] = [2e^{2z}]_{z=-1} = 2e^{-2}$.

Q52. Find the residue of $f(z) = \frac{z}{(z-1)^2}$ at $z = 1$.

Ans: $z = 1$ is a pole of order 2. $\text{Res} = \lim_{z \rightarrow 1} \frac{d}{dz} [z] = 1$.

Q53. Find the residue of $f(z) = \frac{z}{(z+1)(z-2)^2}$ at $z = 2$.

Ans: $z = 2$ is a pole of order 2. $\text{Res} = \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{z}{z+1} \right] = \left[\frac{1}{(z+1)^2} \right]_{z=2} = \frac{1}{9}$.

Q54. State Cauchy's residue theorem.

Ans: If $f(z)$ is analytic inside and on a simple closed curve C except at a finite number of singular points $z_1, \dots, z_n$ inside C , then $\oint_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$.

Q55. Write the analytic part and principal part of a Laurent series.

Ans: For $f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$, the first sum (non-negative powers) is the analytic (regular) part and the second sum (negative powers) is the principal part.

Q56. Find the zeros and poles of $f(z) = \frac{z(z-1)}{z^2+1}$.

Ans: Zeros (numerator = 0): $z = 0, z = 1$ (simple zeros). Poles ($z^2 + 1 = 0$): $z = i, z = -i$ (simple poles).

Q57. Find the zeros of $\frac{z^3 - 1}{z^3 + 1}$.

Ans: Zeros occur where $z^3 - 1 = 0$, i.e. the cube roots of unity $z = 1, e^{2\pi i/3}, e^{-2\pi i/3}$. (The poles are the roots of $z^3 + 1 = 0$.)

Q58. Where is the function $\frac{z^2 - 4}{z^2 + 1}$ not analytic?

Ans: The function fails to be analytic where the denominator vanishes: $z^2 + 1 = 0 \Rightarrow z = i$ and $z = -i$ (simple poles).

Q59. Expand $(z - 1) \sin\left(\frac{1}{z}\right)$ in a Laurent series about $z = 0$ and identify the singularity.

Ans: $\sin \frac{1}{z} = \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \dots$, so $(z - 1) \sin \frac{1}{z} = 1 - \frac{1}{z} - \frac{1}{6z^2} + \frac{1}{6z^3} + \dots$. The principal part has infinitely many terms, so $z = 0$ is an essential singularity.

Q60. Find the residue of $f(z) = \frac{1}{z^2 + 1}$ at $z = i$.

Ans: $z = i$ is a simple pole. $\text{Res} = \lim_{z \rightarrow i} \frac{z - i}{(z - i)(z + i)} = \frac{1}{2i} = -\frac{i}{2}$.

UNIT IV – FOURIER SERIES

Dirichlet's conditions, Euler's formulae, even/odd functions, half-range series and harmonic analysis

Q61. State Dirichlet's conditions for the existence of a Fourier series.

Ans: A function $f(x)$ can be expanded as a Fourier series in an interval of length 2π (or 2ℓ) if: (i) $f(x)$ is single-valued and bounded; (ii) $f(x)$ has at most a finite number of finite discontinuities; (iii) $f(x)$ has at most a finite number of maxima and minima in the interval.

Q62. Write the Fourier series of $f(x)$ in the interval $(-\pi, \pi)$.

Ans: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$, where a_0, a_n, b_n are the Fourier coefficients.

Q63. State Euler's formulae for the Fourier coefficients in $(-\pi, \pi)$.

Ans: $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$, $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$, $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$.

Q64. Write Euler's formulae for a Fourier series in the general interval $(c, c + 2\ell)$.

Ans: $a_0 = \frac{1}{\ell} \int_c^{c+2\ell} f(x) dx$, $a_n = \frac{1}{\ell} \int_c^{c+2\ell} f(x) \cos \frac{n\pi x}{\ell} dx$, $b_n = \frac{1}{\ell} \int_c^{c+2\ell} f(x) \sin \frac{n\pi x}{\ell} dx$.

Q65. Define even and odd functions with examples.

Ans: $f(x)$ is even if $f(-x) = f(x)$ (e.g. $\cos x, x^2, |x|$); $f(x)$ is odd if $f(-x) = -f(x)$ (e.g. $\sin x, x, x^3$).

Q66. What is the Fourier series of an even function in $(-\pi, \pi)$?

Ans: An even function has $b_n = 0$, so its series contains only cosine terms: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$, with $a_0 = \frac{2}{\pi} \int_0^{\pi} f dx$, $a_n = \frac{2}{\pi} \int_0^{\pi} f \cos nx dx$.

Q67. What is the Fourier series of an odd function in $(-\pi, \pi)$?

Ans: An odd function has $a_0 = a_n = 0$, so its series contains only sine terms: $f(x) = \sum_{n=1}^{\infty} b_n \sin nx$,
with $b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$.

Q68. Write the half-range cosine series of $f(x)$ in $(0, \ell)$.

Ans: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\ell}$, where $a_0 = \frac{2}{\ell} \int_0^{\ell} f(x) \, dx$ and $a_n = \frac{2}{\ell} \int_0^{\ell} f(x) \cos \frac{n\pi x}{\ell} \, dx$.

Q69. Write the half-range sine series of $f(x)$ in $(0, \ell)$.

Ans: $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\ell}$, where $b_n = \frac{2}{\ell} \int_0^{\ell} f(x) \sin \frac{n\pi x}{\ell} \, dx$.

Q70. Find a_0 in the Fourier expansion of $f(x) = x$ in $(0, 2\pi)$.

Ans: $a_0 = \frac{1}{\pi} \int_0^{2\pi} x \, dx = \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{2\pi} = \frac{1}{\pi} \cdot 2\pi^2 = 2\pi$.

Q71. Find the constant term $\frac{a_0}{2}$ in the Fourier expansion of $f(x) = x^2$ in $(-\pi, \pi)$.

Ans: $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \, dx = \frac{2}{\pi} \int_0^{\pi} x^2 \, dx = \frac{2\pi^2}{3}$, so the constant term $\frac{a_0}{2} = \frac{\pi^2}{3}$.

Q72. State Parseval's identity (theorem) for Fourier series in $(-\pi, \pi)$.

Ans: If $f(x)$ has the Fourier coefficients a_0, a_n, b_n , then $\frac{1}{\pi} \int_{-\pi}^{\pi} [f(x)]^2 \, dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$.

Q73. Define the root-mean-square (RMS) value of a function.

Ans: The RMS value of $f(x)$ over (a, b) is $\bar{y} = \sqrt{\frac{1}{b-a} \int_a^b [f(x)]^2 \, dx}$.

Q74. Find the RMS value of $f(x) = x$ in $(0, \ell)$.

Ans: $\bar{y} = \sqrt{\frac{1}{\ell} \int_0^{\ell} x^2 \, dx} = \sqrt{\frac{1}{\ell} \cdot \frac{\ell^3}{3}} = \frac{\ell}{\sqrt{3}}$.

Q75. State the complex (exponential) form of the Fourier series in $(-\pi, \pi)$.

Ans: $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$. (In $(-\ell, \ell)$: $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/\ell}$.)

Q76. Write the formula for the complex Fourier coefficient c_n in $(-\pi, \pi)$.

Ans: $c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} \, dx$. (In $(-\ell, \ell)$: $c_n = \frac{1}{2\ell} \int_{-\ell}^{\ell} f(x) e^{-in\pi x/\ell} \, dx$.)

Q77. Define harmonic analysis.

Ans: Harmonic analysis is the process of determining the Fourier coefficients a_0, a_n, b_n of a function specified by a set of tabulated (numerical) values, usually by numerical integration. The term $a_n \cos nx + b_n \sin nx$ is called the n^{th} harmonic.

Q78. Without evaluating the integrals, write a_0, a_n, b_n for $f(x) = x^3$ in $(-\pi, \pi)$.

Ans: Since x^3 is an odd function, $a_0 = 0$ and $a_n = 0$; only the sine coefficients b_n are non-zero.

Q79. To what value does a Fourier series converge at a point of discontinuity?

Ans: At a point of finite (jump) discontinuity x_0 , the Fourier series converges to the average of the left- and right-hand limits, $\frac{1}{2}[f(x_0^-) + f(x_0^+)]$.

Q80. Find b_n in the Fourier series of $f(x) = x$ in $(-\pi, \pi)$.

Ans: As x is odd, $b_n = \frac{2}{\pi} \int_0^\pi x \sin nx \, dx = \frac{2}{\pi} \cdot \frac{\pi(-1)^{n+1}}{n} = \frac{2(-1)^{n+1}}{n}$.
